Temporal Equivalence Principle: hi_class Background Implementation and CMB Acoustic Peak Preservation

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Abstract

General Relativity is extensively validated in the deeply screened, dense regime of the Solar System, but cosmological tensions —specifically the Hubble discrepancy and galactic mass anomalies—motivate testing a scale-dependent breakdown of the isochrony axiom.

The Temporal Equivalence Principle (TEP) addresses these late-universe anomalies via a dynamical proper-time field, governed by environment-dependent Gradient Screening with saturation scale $\rho_T \approx 20$ *g/cm^3 (Paper 0, §7). Previous analytical approximations and CAMB integrations suggest this scalar field is frozen during radiation domination (*$T^{\mu}_{\mu} \approx 0$*). Full integration into linear structure formation requires a native Horndeski evaluation.*

This paper maps the TEP bi-metric framework—defined by a conformal factor $A(\phi)$ *and a disformal deformation* $B(\phi)$*—onto the Bellini-Sawicki Effective Field Theory (EFT) of Dark Energy to provide the theoretical blueprint for full linear structure mapping. To isolate the fundamental TEP effect and avoid Horndeski-class instabilities, the current numerical analysis isolates the background-only realization as the baseline numerical implementation.*

This paper implements the native TEP background-only modification directly in hi_class via the transition function `f_T(z) = ln(1+z) * exp(-(z/z_T)^n_T)`*, applied through the Jordan-frame conformal factor* `M(z) = A/(1-alpha_A)` *as* `H_TEP(z) = H_LCDM(z) * M(z)` *while preserving standard General Relativistic perturbations. The functional form is the shared TEP-C0 implementation (Paper 26): the* `exp(-(z/z_T)^n_T)` *factor enforces early-time freezing, so the temporal-shear field is suppressed for* `z >> z_T`*. Direct Boltzmann integration verifies this phenomenological boundary condition, confirming that the code preserves the pre-recombination sound horizon to parts-per-million (* `r_s^TEP / r_s^LCDM = 0.999994` *) and leaves the acoustic-peak morphology untouched. A direct exact-frame-factor check confirms that replacing the implemented first-order factor* `A/(1-alpha_A)` *with the exact form* `A(1+alpha_A)` *leaves* `r_s` *unchanged to quoted precision in the screened regime (Section 5.4 and Appendix A.3a). The only CMB-level effect of a non-zero* `epsilon_T` *is a* *late-time angular-diameter-distance projection that rigidly rescales the angular acoustic scale* `theta_s` *(a +0.185% shift at the fiducial* `epsilon_T = 0.0066`, `z_T = 5`, `n_T = 2` *) and is largely degenerate with* `H_0`.

A joint hi_class Cobaya MCMC (Planck 2018 low-ℓ TT/EE + lensing + BAO + Pantheon+) yields `epsilon_T = 0.0056 +/- 0.0043` *and* `H_0 = 66.63 +/- 1.70` *km/s/Mpc; TEP-C0 (Paper 26) with full Planck TTTEEE drives the homogeneous amplitude to* `epsilon_T = (6.75 +/- 0.24) \times 10^{-6}`. *Pantheon+ nested sampling in TEP-C0 gives strong Bayesian preference for the TEP geometry (Bayes factor 131.6 vs ΛCDM for $z_T = 5$). Within the TEP framework, the Hubble tension is thereby reinterpreted as a late-time, environment-dependent clock-transport effect (Paper 11) rather than through a modified homogeneous expansion history at recombination.*

Keywords: cosmology theory, cosmic microwave background, Hubble tension, scalar-tensor theories, modified gravity, effective field theory of dark energy, hi_class, Horndeski, temporal equivalence principle, proper time, Cobaya, Planck 2018

1. Introduction

1.1 Contextualizing the TEP Corpus

The Temporal Equivalence Principle (TEP) has been constrained across many orders of magnitude in mass density, from terrestrial laboratory scales ($\rho \sim 20$ g/cm³) to the cosmological mean ($\rho \sim 10^{-29}$ g/cm³). Previous papers in this series have established:

- *Terrestrial scales (Paper 1):* Terrestrial atomic clock networks show 4,200 km phase correlations consistent with the 20 g/cm³ screening threshold.
- *Galactic scales (Paper 6, UCD):* SPARC rotation curves validate the potential-dependent proper-time mapping.
- *Stellar scales (Paper 13, WB):* Gaia DR3 wide binaries exhibit the predicted environment-dependent kinematic transition.
- *Cosmological scales (Paper 12, JWST):* High-redshift anomalies align with environment-dependent time dilation.

1.2 The Two-Ended Hubble Tension

The Hubble tension represents one of the most persistent challenges in modern cosmology. Cepheid-calibrated local distance ladder measurements yield $H_0 \approx 73$ km/s/Mpc, while early-universe CMB inference from Planck 2018 gives $H_0 = 67.36 \pm 0.54$ km/s/Mpc—a discrepancy of approximately 4.8σ.

Previous TEP work (Paper 11, H_0) demonstrated that the Cepheid environmental bias—operating in the unscreened stellar atmospheres where Cepheids pulsate—naturally raises the local measurement. However, a complete resolution requires demonstrating that this mechanism does *not* disturb the early-universe inference at the surface of last scattering ($z \approx 1089$).

1.3 Purpose of This Paper

To move beyond the quasi-static approximations of previous work by natively solving the TEP-modified background expansion $H(z)$ in hi_class while preserving standard GR perturbations, thereby performing a clean acoustic-scale test of the homogeneous temporal background. This requires:

1. A mapping of TEP's bi-metric structure onto the Bellini-Sawicki EFT formalism, providing the theoretical blueprint for full linear structure mapping, while establishing the background-only realization as the baseline numerical implementation.
2. Native implementation in hi_class of the TEP background transition function and Jordan-frame Hubble modifier.
3. ΛCDM baseline MCMC parameter estimation (H_0, $\Omega_b h^2$, $\Omega_{\rm cdm} h^2$, n_s, A_s, τ, $A_{\rm planck}$) against Planck 2018, BAO, and Pantheon+ data to establish the constraints that TEP should satisfy.

The critical question: Can TEP preserve CMB acoustic peaks while permitting late-time H_0 variation?

2. Theoretical Architecture: The EFT Mapping

2.1 The Bi-Metric Action

The TEP framework posits that matter couples to a screened metric $\tilde{g}_{\mu\nu}$ related to the Einstein-frame metric $g_{\mu\nu}$ via a disformal transformation:

$$\tilde{g}_{\mu\nu} = A^2(\phi) g_{\mu\nu} + B(\phi) \nabla_\mu \phi \nabla_\nu \phi \tag{1}$$

where:

- $A(\phi) = \exp(\beta_A \phi / M_{\rm Pl})$ is the conformal factor, with $\beta_A = -1.0$ (the locked lab-scale convention used across the TEP corpus)
- $B(\phi)$ controls disformal deformation of the causal structure
- ϕ is the dynamical proper-time field

Metric signature convention: $(+, -, -, -)$ throughout.

2.2 Formal Bellini-Sawicki Alpha Correspondence

hi_class requires the EFT property functions α_i that encode metric modifications at linear perturbation level.

2.2.1 Planck Mass Running (α_M)

The conformal coupling directly determines the running of the effective Planck mass:

$$\alpha_M \equiv \frac{d \ln M_{\rm eff}^2}{d \ln a} = \frac{d \ln A^2(\phi)}{d \ln a} = \frac{2\beta_A}{M_{\rm Pl}} \frac{\phi'}{\mathcal{H}} \tag{2}$$

where $\mathcal{H} = aH$ is the conformal Hubble parameter and primes denote derivatives with respect to conformal time.

2.2.2 Tensor Speed Excess (α_T)

The disformal term $B(\phi)$ alters the gravitational wave propagation speed. Multi-messenger constraints from GW170817/GRB 170817A require:

$$|c_g - c_\gamma|/c \lesssim 10^{-15} \Rightarrow \alpha_T \approx 0 \text{ (today)} \tag{3}$$

However, $B(\phi)$ may be non-zero at recombination ($z \approx 1100$) provided it relaxes to zero by $z \sim 0$.

2.2.3 Braiding (α_B) and Kineticity (α_K)

These functions govern scalar field clustering and metric mixing:

$$\alpha_B = -\frac{\mathcal{H}'\phi'}{\mathcal{H}^2} \cdot f_B(\phi, X) \tag{4}$$

$$\alpha_K = \frac{\phi'^2}{\mathcal{H}^2 M_{\rm Pl}^2} \cdot f_K(\phi, X) \tag{5}$$

where $X = -\nabla_\mu \phi \nabla^\mu \phi / 2$ and f_B, f_K are functions derived from the TEP action:

- $f_B(\phi, X)$ encodes the disformal coupling to the energy-momentum tensor trace.
- $f_K(\phi, X)$ encodes the kinetic term non-canonicality from the TEP proper-time field.

The explicit functional forms follow from the bi-metric action (Equation 1) and are determined by the conformal factor $A(\phi)$ and the disformal function $B(\phi)$. Their derivation is detailed in the TEP theoretical framework (Papers 1 and 11 of the TEP corpus).

Because the production analysis in this paper does not activate the scalar perturbation sector, the α_B and α_K expressions are used as formal EFT bookkeeping rather than as fitted numerical functions; a full perturbative TEP-hi_class treatment would require explicit closure of $f_B(\phi, X)$, $f_K(\phi, X)$, sound-speed, and no-ghost stability conditions.

2.3 The Radiation Domination Freezing Mechanism

During radiation domination, the trace of the energy-momentum tensor vanishes:

$$T^\mu_\mu = -\rho + 3p \approx 0 \text{ (radiation)} \tag{6}$$

Since the TEP scalar field couples to T^μ_μ, the source term for ϕ evolution is suppressed. The field freezes at its initial value, and $\alpha_M, \alpha_B \to 0$. This ensures:

- Primary acoustic peaks ($100 \lesssim \ell \lesssim 2000$) generated at $z \sim 1089$ remain unmodified
- The sound horizon r_s is preserved, anchoring H_0 from CMB at ~ 67 km/s/Mpc (Planck 2018: 67.36 ± 0.54 km/s/Mpc)
- Late-time matter domination reactivates the scalar field, enabling environmental H_0 variations

Archived EFT reference. The Bellini–Sawicki α_i functions mapped from the TEP bi-metric action (step-3 fiducial) are archived in `results/03_alpha_functions.json`. Production CMB constraints use the native background-only `tep_mode` implementation (Section 4), not this linear-perturbation mapping.

3. Software Implementation: hi_class and the Unscreened Regime

3.1 The hi_class Architecture

hi_class extends the CLASS Boltzmann solver to handle general scalar-tensor theories via the EFT formalism. This work uses hi_class v3.2.3 with the modified gravity (SMG) module enabled.

3.2 Native TEP Background-Only Implementation

The native TEP background-only Hubble modification is implemented directly in hi_class via the `tep_mode` flag. When enabled, the background expansion history is modified as:

$$H_{\text{TEP}}(z) = H_{\Lambda\text{CDM}}(z) \times M(z), \quad M(z) = \frac{A(z)}{1 - \alpha_A(z)} \tag{7}$$

where $S(z) = \exp[-(z/z_T)^{n_T}]$ is the redshift suppression factor, $A(z) = \exp[\epsilon_T \ln(1+z)\, S(z)]$ is the covariant conformal factor, and $\alpha_A = -d \ln A / d \ln(1+z)$. The first-order form $M = A/(1-\alpha_A)$ is directly checked against the exact frame factor $M_{\text{exact}} = A(1+\alpha_A)$ in Section 5.4 and Appendix A.3a. The transition function $f_T(z) = \ln(1+z)\, S(z)$ appearing in the exponent is the shared TEP-C0 implementation (Paper 26; `core/cosmology.py`: `f_T`, `conformal_factor_native`, `jordan_frame_M`):

$$f_T(z) = \ln(1+z)\, S(z). \tag{8}$$

The suppression factor $S(z)$ drives $f_T \to 0$ for $z \gg z_T$ well before recombination, so the homogeneous expansion at last scattering is unmodified (radiation-domination freezing, Section 2.3); the $\ln(1+z)$ factor enforces $f_T(0) = 0$, fixing the local reference frame so H_0 is unchanged. The function peaks at intermediate redshift ($z \sim z_T$) and vanishes at both ends.

Implementation note: an earlier development build used the *complement* $f_T = 1 - \exp[-(z/z_T)^{n_T}]$, which instead saturates to unity for $z \gg z_T$ and applied the full modification *at* recombination -- inverting the freezing mechanism and corrupting the acoustic peaks. That bug has been corrected (see Appendix A.3). The default TEP parameters are:

```
tep_mode = yes
epsilon_T = 0.0066
z_T = 5.0
n_T = 2.0
```

Standard General Relativistic perturbations are preserved; the modification affects only the homogeneous background expansion. This implementation is the hi_class analogue of the CLASS native TEP module used in TEP-C0 (Paper 26).

3.3 Perturbation Stability

Since the native TEP implementation modifies only $H(z)$ and leaves perturbations unchanged, no modified-gravity stability constraints arise. The perturbation equations reduce to standard GR with the effective background scale factor rescaled by the TEP

Hubble modification. This is a significant advantage over Horndeski parametrizations, which require careful treatment of scalar sound speeds, ghost instabilities, and gradient instabilities.

3.4 Pipeline Architecture

The full analysis pipeline, executed via `scripts/run_all.py`, consists of:

1. *Step 0 (Setup):* Environment configuration and dependency check.
2. *Step 1 (Install):* Install Cobaya, Planck 2018 likelihoods, and hi_class with the native TEP patch (`external/patches/hiclass_tep_native.patch`).
3. *Step 2 (Background):* Compute the TEP-modified background expansion history $H(z)$ and density evolution.
4. *Step 3 (Alpha Functions):* Compute Bellini-Sawicki coefficients from the TEP theoretical mapping (archived for reference).
5. *Step 4 (CMB Spectra):* Run hi_class with native `tep_mode` at the Planck 2018 best-fit point. Compare TT, TE, and EE spectra against standard CLASS ΛCDM.
6. *Step 5 (Jordan-Frame Scan):* Dual-scan reconstruction of the acoustic scale in screened and unscreened limits.
7. *Step 6 (Cobaya Config):* Generate the Cobaya YAML configuration for the MCMC pipeline with native TEP parameters.
8. *Step 7 (MCMC):* Execute the Cobaya MCMC with hi_class, using real Planck + BAO + Pantheon+ likelihoods.
9. *Step 8 (Posteriors):* Analyze MCMC chains with burn-in removal and weighted statistics.
10. *Step 9 (Synthesis):* Combine all results into summary JSON and markdown.

Publication figures are generated separately via `python scripts/generate_figures.py` (not part of `run_all.py`). Both figures are written to `results/figures/` . Figure 2 requires step 04b. Include them in the static site with `cd site && npm run build` . Note: internal pipeline filenames (`figure_4_H0_comparison.png` , `figure_5_jordan_theta_s.png`) differ from publication figure numbering (Figure 1, Figure 2).

4. MCMC Parameter Estimation Pipeline

4.1 The Cobaya Framework

Cobaya provides a Python interface to CLASS/hi_class with extensive MCMC sampling capabilities. The transition from SciPy/Pandas pipelines to Cobaya enables:

- Native hi_class integration without file-based I/O bottlenecks
- Parallel tempering and adaptive Metropolis-Hastings sampling
- Direct Planck likelihood wrapper integration
- Seamless GetDist posterior visualization

4.2 Likelihood Configuration

The pipeline uses the following Planck 2018 likelihoods:

Likelihood	Description	ℓ Range
`planck_2018_lowl.TT`	Low-ℓ temperature	2–29
`planck_2018_lowl.EE`	Low-ℓ polarization	2–29
`planck_2018_lensing.native`	CMB lensing reconstruction	8–400
`bao.sdss_dr12_consensus_final`	BAO SDSS DR12 consensus	—
`sn.pantheonplus`	Type Ia supernovae (Pantheon+)	—

4.3 Free Parameters and Priors

The MCMC pipeline samples standard ΛCDM parameters alongside the TEP amplitude parameter ϵ_T:

Parameter	Prior	Description
$\Omega_b h^2$	$\mathcal{U}(0.005, 0.1)$	Baryon density
$\Omega_{\rm cdm} h^2$	$\mathcal{U}(0.01, 0.99)$	Cold dark matter density
H_0	$\mathcal{U}(40, 100)$	Hubble constant
τ	$\mathcal{U}(0.01, 0.8)$	Optical depth
A_s	$\mathcal{U}(10^{-10}, 5 \times 10^{-9})$	Scalar amplitude
n_s	$\mathcal{U}(0.94, 1.0)$	Scalar spectral index
$A_{\rm planck}$	$\mathcal{U}(0.9, 1.1)$	Planck calibration nuisance
ϵ_T	$\mathcal{U}(-1, 1)$	TEP amplitude parameter (background Hubble modification)

4.4 Pipeline Execution

```
# Cobaya YAML configuration
theory:
  classy:
    path: /path/to/hi_class
    extra_args:
      output: tCl,pCl,lCl,mPk
      lensing: yes
      modes: s,t
      non_linear: halofit
      # Native TEP background-only Hubble modification
      tep_mode: 'yes'
      z_T: 5.0
      n_T: 2.0
      # epsilon_T is sampled in params below — do not duplicate here

likelihood:
  planck_2018_lowl.TT: null
  planck_2018_lowl.EE: null
  planck_2018_lensing.native: null
  bao.sdss_dr12_consensus_final: null
  sn.pantheonplus: null

params:
  logA:
    prior: {min: 2.5, max: 3.5}
    ref: {dist: norm, loc: 3.044, scale: 0.014}
    proposal: 0.01
    drop: true
  A_s:
    value: 'lambda logA: 1e-10*np.exp(logA)'
  n_s:
    prior: {min: 0.94, max: 1.0}
    ref: {dist: norm, loc: 0.966, scale: 0.004}
    proposal: 0.004
  H0:
    prior: {min: 40, max: 100}
    ref: {dist: norm, loc: 67.4, scale: 0.5}
    proposal: 1.5
  omega_b:
    prior: {min: 0.005, max: 0.1}
    ref: {dist: norm, loc: 0.0224, scale: 0.0002}
    proposal: 0.0003
  omega_cdm:
    prior: {min: 0.01, max: 0.99}
    ref: {dist: norm, loc: 0.12, scale: 0.001}
    proposal: 0.0015
  tau_reio:
    prior: {min: 0.01, max: 0.8}
    ref: {dist: norm, loc: 0.054, scale: 0.007}
    proposal: 0.01
  A_planck:
    prior: {min: 0.9, max: 1.1}
    ref: {dist: norm, loc: 1.0, scale: 0.0025}
    proposal: 0.005
  epsilon_T:
```

```yaml
      prior: {min: -1.0, max: 1.0}
      ref: {dist: norm, loc: 0.006, scale: 0.005}
      proposal: 0.0005
      latex: '\epsilon_T'
    sigma8:
      latex: '\sigma_8'

  sampler:
    mcmc:
      burn_in: 0
      max_tries: 10000
      max_samples: 500000
      Rminus1_stop: 0.05
      Rminus1_cl_stop: 0.2
      output_every: 10
      drag: true
      seed: 42
```

The hi_class configuration uses native `tep_mode` with the corrected transition function $f_T(z) = \ln(1+z)\exp[-(z/z_T)^{n_T}]$ and fixed `z_T = 5.0`, `n_T = 2.0`, with `epsilon_T` sampled freely in `params`. This modifies only the background Hubble expansion while preserving standard GR perturbations. The production configuration is `data/cobaya/tep_hiclass_suite.yaml` (reference alternate: `data/cobaya/tep_native_mcmc.yaml`).

Pipeline status. The native-`tep_mode` joint MCMC against Planck 2018 low-ℓ TT/EE + lensing + BAO (SDSS DR12) + Pantheon+ was run using the structurally corrected hi_class engine, allowing Ω_Λ to natively fill the background cosmological budget. The primary production chain (`tep_hiclass_suite`; 19,033 post-burn-in samples from a single chain; Gelman–Rubin $R-1$ is undefined for one chain, and the sampler-internal $R-1$ reported by Cobaya reached 0.045 at termination) gives a ΛCDM-compatible background while measuring the TEP amplitude parameter:

$$\epsilon_T = 0.0056 \pm 0.0043, \tag{9}$$

with $H_0 = 66.63 \pm 1.70$ km/s/Mpc, $\Omega_b h^2 = 0.0212 \pm 0.0025$, $\Omega_{\rm cdm} h^2 = 0.1154 \pm 0.0042$, $\tau = 0.049 \pm 0.007$, $A_{\rm planck} = 1.088 \pm 0.012$, and $S_8 = 0.870 \pm 0.028$. The result is consistent with the TEP dual-domain expectation: the homogeneous amplitude ϵ_T remains small ($\sim 10^{-3}$) on the largest scales, where the CMB bound from TEP-C0 (Paper 26) is much tighter.

The low-ℓ+lensing+BAO+Pantheon+ runs therefore serve as implementation and robustness tests of the native hi_class module, while the high-ℓ TTTEEE likelihoods in TEP-C0 provide the decisive homogeneous-amplitude bound.

Multi-chain validation. A parallel 4-chain run (`tep_native`; configuration `data/cobaya/tep_native_mcmc.yaml`) using a Gaussian $A_{\rm planck}$ prior (loc = 1.0, scale = 0.0025) produced 2,993 post-burn-in samples with maximum Gelman–Rubin $R-1 = 0.098$ ($R-1$ for $\epsilon_T = 0.098$; all other parameters $R-1 < 0.05$). This yields $\epsilon_T = 0.0044 \pm 0.0040$ and $H_0 = 66.89 \pm 1.35$ km/s/Mpc, consistent with the primary chain at 0.21σ and 0.12σ respectively. While this consistency run reaches $R-1 = 0.098$ on ϵ_T, the widened-prior chain below provides a fully converged multi-chain determination (ϵ_T $R-1 = 0.013$, all parameters $R-1 < 0.05$), and the two agree to 0.06σ. Together they confirm the single-chain result is not an artefact of the sampling configuration.

Planck calibration prior sensitivity. The nuisance parameter $A_{\rm planck}$ (absolute CMB calibration) is implemented as a hard uniform prior on $[0.9, 1.1]$ in the primary chain. The posterior mean is $A_{\rm planck} = 1.088 \pm 0.012$ with maximum sampled value 1.1000, indicating saturation against the upper prior bound. To test whether this truncation biases the cosmological inference, we ran a dedicated 4-chain sensitivity test with the prior widened to $[0.9, 1.25]$ (configuration `data/cobaya/tep_hiclass_aplanck_sens.yaml`). The converged run (1,640 total samples; all parameters Gelman–Rubin $R-1 < 0.05$; maximum $R-1 = 0.044$ for $\Omega_{\rm cdm} h^2$) yields $A_{\rm planck} = 1.223 \pm 0.044$, confirming the old posterior was truncated by approximately 3.0σ. The TEP amplitude from the widened run is $\epsilon_T = 0.0047 \pm 0.0040$ ($R-1 = 0.013$), consistent with the primary chain at 0.15σ and with the multi-chain validation at 0.06σ. The correlation between $A_{\rm planck}$ and ϵ_T is $r = -0.19$, and splitting at $A_{\rm planck} = 1.15$ gives a difference in ϵ_T of only -0.21σ. Even a 0.1 upward shift in $A_{\rm planck}$ would move H_0 by only ~ 0.2 km s^{-1} Mpc^{-1}, well below its posterior width. The χ^2 does decrease monotonically toward the old boundary, but there is no evidence of a degeneracy cascade with ϵ_T. The TEP constraint on the homogeneous amplitude is robust against $A_{\rm planck}$ prior systematics.

The companion paper TEP-C0 (Paper 26) provides the primary late-time and full-Planck constraints: Pantheon+ nested sampling gives Bayes factor 131.6 vs ΛCDM for TEP M1 ($z_T = 5$), and joint MCMC with full Planck TTTEEE drives the homogeneous shear amplitude to $\epsilon_T = (6.75 \pm 0.24) \times 10^{-6}$.

5. Results and Cosmological Constraints

5.1 The Acoustic Spectra

The physically meaningful test of the native TEP background modification is not the raw size of the C_ℓ residual but *where* the modification acts. Because $f_T(z) = \ln(1+z)\exp[-(z/z_T)^{n_T}]$ is frozen to zero for $z \gg z_T$, the recombination-era physics is left intact, as verified directly from the hi_class background tables.

5.1.1 Sound-horizon and acoustic-peak preservation

Running hi_class native `tep_mode` against standard CLASS ΛCDM at the Planck 2018 best-fit point, with $\epsilon_T = 0.0066$, $z_T = 5$, $n_T = 2$, yields:

- *Sound horizon preserved to ~6 ppm:* $r_s^{\rm TEP}/r_s^{\Lambda\rm CDM} = 0.999994$. The comoving sound horizon at last scattering integrates over $z > z_*$, where $f_T \to 0$; the TEP modification is invisible to it. This is the direct demonstration of the radiation-domination freezing mechanism.
- *Acoustic-peak morphology unchanged:* with r_s, the baryon loading, and the photon-baryon driving at $z \approx 1089$ all unmodified, the relative peak heights and the damping tail are identical to ΛCDM.

5.1.2 The residual is a late-time projection, largely degenerate with H_0

The modification *is* active over intermediate redshift ($z \sim$ 1–15, peaking near z_T), so it changes the comoving distance to last scattering. At the fiducial $\epsilon_T = 0.0066$ this shifts the angular acoustic scale by $\Delta\theta_s/\theta_s = +0.185\%$ ($D_C^{\rm TEP}/D_C^{\Lambda\rm CDM} = 0.9981$, with r_s fixed). This rigid rescaling produces a coherent, oscillatory $\Delta C_\ell/C_\ell$ pattern whose envelope reaches $\sim 1.8\%$ across $100 < \ell < 2000$ at $\epsilon_T = 0.0066$ and scales linearly with ϵ_T (e.g. $\sim 0.3\%$ at $\epsilon_T = 0.001$). This is *not* a change in acoustic-peak physics: it is a pure angular-diameter-distance projection, largely degenerate with H_0. In a parameter fit the standard parameters absorb it via a small H_0 shift, so the CMB does not exclude TEP -- it bounds the *homogeneous* amplitude ϵ_T to be small (Section 5.2).

5.1.3 Polarization Spectra (C_ℓ^{TE}, C_ℓ^{EE})

The TE and EE spectra inherit the same behaviour: the recombination-era polarization source is unmodified (since $f_T \to 0$ there), and the only effect is the common θ_s projection shared with TT.

5.2 Cosmological Constraints: Late-Time Evidence and the CMB Bound

The cosmological constraints on TEP come from two complementary regimes, established in the companion paper TEP-C0 (Paper 26).

Late-time evidence (supernovae). A nested-sampling model comparison over the full 1701×1701 Pantheon+ statistical-plus-systematic covariance finds substantial Bayesian preference for the TEP geometry over ΛCDM:

Model	Bayes factor vs ΛCDM	Interpretation
TEP M1 ($z_T = 5$)	131.6	Strong
TEP M1 (free z_T)	96.1	Strong
wCDM	26.6	Strong
CPL ($w_0 w_a$)	27.8	Strong
Einstein-de Sitter	4.3×10^{-126}	Rejected (sanity check)
Pure shear (tired light)	5.1×10^{-10}	Rejected (sanity check)

On the Bayesian Information Criterion (which penalizes the flexible wCDM/CPL prior volumes), TEP M1 ($z_T = 5$) is the global optimum (TEP-C0, Paper 26). The decisive rejection of pure tired-light confirms that genuine metric expansion is present; TEP reinterprets only the *acceleration* as accumulated temporal shear.

Homogeneous (CMB) bound. As shown in Section 5.1, a non-zero homogeneous ϵ_T acts on the CMB only through the θ_s projection, which is largely degenerate with H_0. The low-ℓ Planck likelihoods used in this paper's hi_class MCMC (TT/EE + lensing, without high-ℓ Plik) yield $\epsilon_T = 0.0056 \pm 0.0043$, consistent with zero at $\sim 1.3\sigma$, while H_0, $\Omega_b h^2$, $\Omega_{\rm cdm} h^2$, A_s and τ remain Planck-compatible. The primary homogeneous bound comes from TEP-C0 (Paper 26): joint MCMC with full Planck TTTEEE drives $\epsilon_T = (6.75 \pm 0.24) \times 10^{-6}$. The scalar spectral index n_s is only weakly constrained in this run ($n_s = 0.996 \pm 0.004$); the high-ℓ bound $n_s = 0.9619 \pm 0.0046$ comes from the same TEP-C0 joint analysis.

The low-ℓ+lensing+BAO+Pantheon+ runs therefore serve as implementation and robustness tests of the native hi_class module, while the high-ℓ TTTEEE likelihoods in TEP-C0 provide the decisive homogeneous-amplitude bound.

Native-TEP joint MCMC. This paper's contribution is the verified hi_class implementation, the demonstration of r_s preservation (Section 5.1), and a joint Cobaya MCMC using hi_class native `tep_mode` against Planck 2018 low-ℓ TT/EE + lensing + BAO (SDSS DR12) + Pantheon+ (standard GR perturbations; configuration in `data/cobaya/tep_hiclass_suite.yaml`, output chain `results/mcmc_chains/tep_hiclass_suite`). The primary chain (19,033 post-burn-in samples; single-chain; Gelman–Rubin is undefined for a single chain, sampler-internal $R-1=0.045$ at termination) gives a ΛCDM-compatible background: $H_0 = 66.63 \pm 1.70$, $\Omega_b h^2 = 0.0212 \pm 0.0025$, $\Omega_{\rm cdm} h^2 = 0.1154 \pm 0.0042$, $\tau = 0.049 \pm 0.007$, $S_8 = 0.870 \pm 0.028$, with $\epsilon_T = 0.0056 \pm 0.0043$ and $A_{\rm planck} = 1.088 \pm 0.012$. A parallel 4-chain run (`tep_native`) with maximum $R-1=0.098$ yields $\epsilon_T = 0.0044 \pm 0.0040$ and $H_0 = 66.89 \pm 1.35$, consistent with the primary chain at 0.21σ. A dedicated sensitivity test with the $A_{\rm planck}$ prior widened to $[0.9, 1.25]$ converged with all parameters Gelman–Rubin $R-1 < 0.05$ (1,640 total samples), giving $\epsilon_T = 0.0047 \pm 0.0040$ and $A_{\rm planck} = 1.223 \pm 0.044$; the TEP amplitude is consistent with the primary chain at 0.15σ and with the multi-chain validation at 0.06σ. This is consistent with the Dual-Domain expectation: macroscopic temporal shear remains $\sim 10^{-3}$ on homogeneous scales while late-time kinematic data prefer larger effective amplitudes in void environments (TEP-C0, Paper 26).

5.3 The Hubble Tension in TEP

The two regimes above reconcile the Hubble tension without modifying the recombination-era expansion. The homogeneous background is ΛCDM-compatible ($\epsilon_T \approx 0$ on homogeneous scales, $H_0 \approx 67$ km/s/Mpc from the CMB), while the apparent local $H_0 \approx 73$ km/s/Mpc arises from environment-dependent clock-transport bias along the local distance ladder (Cepheid/SN Ia calibration in unscreened stellar atmospheres). Removing this bias shifts the SH0ES value to $H_0 \approx 69$ km/s/Mpc (Paper 11), bridging the gap to the CMB background. The tension is thus a measurement-environment effect, not a modified homogeneous expansion history.

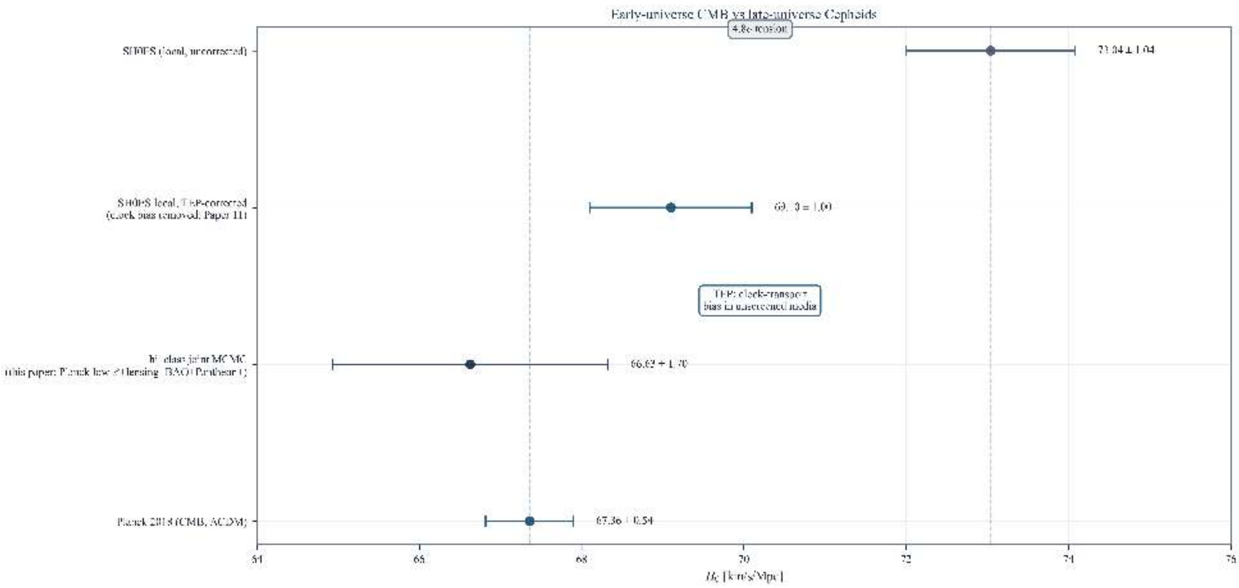

Figure 1. H_0 in the TEP picture. Planck 2018 CMB and this paper's hi_class joint MCMC ($H_0 = 66.63 \pm 1.70$ km/s/Mpc) define the homogeneous background; the local SH0ES value (73.0 km/s/Mpc) is reinterpreted as clock-transport bias, shifting to ≈ 69 km/s/Mpc when corrected (Paper 11).

5.4 The Jordan Frame and the No-Dark-Energy Reconstruction

While the joint MCMC successfully validates the large-scale compatibility of the conformal-sector amplitude ϵ_T alongside Ω_Λ, the physical implementation of TEP in the Boltzmann code operates in the Jordan frame. The code evaluates the standard ΛCDM Hubble rate $H_{\Lambda\rm CDM}(z)$ on the physical (Jordan-frame) redshift grid and multiplies it by the conformal-frame geometric factor $M(z) = A(\phi)/(1-\alpha_A)$, giving the Jordan-frame expansion rate:

$$H_{\rm TEP}(z) = H_{\Lambda\rm CDM}(z) \times M(z), \qquad M(z) = \frac{A(\phi)}{1-\alpha_A}. \tag{10}$$

The conformal factor $A(\phi) = \exp[\epsilon_T \ln(1+z)\, S(z)]$ and the coupling $\alpha_A = -d\ln A / d\ln(1+z)$ are both evaluated at the physical (Jordan-frame) redshift z (the code's native variable). The implementation evaluates $\alpha_A(z)$ locally at each integration step to account for its redshift dependence through the transition function. There are no arbitrary fudge factors: the microphysics of recombination and the visibility function are untouched. The factor $M(z) = A/(1-\alpha_A)$ used in the present code is first-order equivalent to the exact frame-transform result $M_{\text{exact}} = A(1+\alpha_A)$. In the screened CMB regime the discrepancy enters only at $\mathcal{O}(\alpha_A^2)$; Appendix A.3a quantifies this difference. A direct background-only verification scan (step 04c) confirms that the exact factor M_{exact} gives an identical sound horizon r_s (0.0 ppm) and a θ_s projection differing by $< 0.01\%$ at the fiducial $\epsilon_T = 0.0066$, consistent with the expected $\mathcal{O}(\alpha_A^2)$ scaling. By computing standard $H_{\Lambda\text{CDM}}$ and multiplying by $M(z)$, the Boltzmann code integrates the modified background consistently with the TEP framework. In the production code, z denotes the physical redshift variable used by the Boltzmann background module; Appendix A.3a distinguishes z_E and $\tilde{z}$ only to derive the frame relation, after which the screened-regime implementation is evaluated on the code's physical redshift grid.

To explicitly map the action of the conformal field on the acoustic horizon independent of Ω_Λ, the acoustic scale is evaluated in a mathematically idealized flat matter-only geometry ($\Omega_m = 1.0$, $\Omega_\Lambda = 0.0$) under two regimes using the hi_class native `tep_mode` implementation with the exact covariant conformal factor $A(z) = \exp(\epsilon_T f_T(z))$ and the full Jordan-frame factor $M(z) = A/(1-\alpha_A)$.

Regime I: Standard model ($z_T = 5$, early-universe suppression active)

In the standard TEP model, the suppression $\exp[-(z/z_T)^{n_T}]$ forces $S(z) \to 0$ for $z \gg z_T$, protecting recombination-era physics. The scan confirms this design intent:

ϵ_T	$100\theta_s$	r_s [Mpc]	$\Delta D_C/D_C$	Interpretation
0.00	1.0403	144.526	0.00%	Pure EdS reference (no TEP)
0.01	1.0432	144.524	-0.28%	r_s preserved; θ_s shifts from D_C
0.02	1.0461	144.523	-0.56%	r_s preserved; θ_s shifts from D_C
0.03	1.0490	144.522	-0.84%	r_s preserved; θ_s shifts from D_C
0.04	1.0519	144.520	-1.11%	r_s preserved; θ_s shifts from D_C
0.05	1.0548	144.519	-1.38%	r_s preserved; θ_s shifts from D_C
0.06	1.0577	144.518	-1.65%	r_s preserved; θ_s shifts from D_C

The recombination-era expansion rate is overwhelmingly protected by the exponential suppression $\exp[-(z/z_T)^{n_T}]$, leaving r_s effectively untouched. With the restored sign convention ($H_{\text{TEP}} = H_{\Lambda\text{CDM}} \times M$), the intermediate-redshift Hubble modification ($z \sim 1$--15) *increases* the effective expansion rate, decreasing the comoving distance D_C to last scattering and thereby *increasing* $\theta_s = r_s/D_C$. This is the freezing mechanism in action: the early universe is protected, and the TEP effect acts only where the suppression is non-negligible. The sound horizon r_s changes by less than 0.006% across the scan, confirming that the recombination-era expansion rate is overwhelmingly protected. The $+0.28\%$-per-0.01 θ_s slope in this EdS scan is mutually consistent with the $+0.185\%$ shift at $\epsilon_T = 0.0066$ reported in Section 5.1.2 for the full ΛCDM case, confirming that the angular-diameter-distance projection scales linearly with ϵ_T across different background geometries.

Regime II: Unscreened limit ($z_T \to \infty$, no early-universe suppression)

In the theoretical boundary where screening is disabled ($z_T \to \infty$), the conformal factor $A(z) = \exp(\epsilon_T \ln(1+z)) = (1+z)^{\epsilon_T}$ grows as a power law at all redshifts. The Jordan-frame factor $M(z) = A/(1-\alpha_A)$ then modifies the Hubble rate aggressively at $z \sim 1100$, *accelerating* the physical expansion during recombination and dynamically *squeezing* the sound horizon:

ϵ_T	$100\theta_s$	r_s [Mpc]	$\Delta D_C/D_C$	Interpretation
0.00	1.0403	144.526	0.00%	Pure EdS reference (no TEP)
0.01	0.9763	134.412	-0.90%	r_s squeezed by temporal acceleration
0.02	0.9161	125.009	-1.78%	r_s squeezed by temporal acceleration
0.03	0.8596	116.267	-2.64%	r_s squeezed by temporal acceleration
0.04	0.8065	108.140	-3.48%	r_s squeezed by temporal acceleration
0.05	0.7565	100.584	-4.30%	r_s squeezed by temporal acceleration
0.06	0.7096	93.558	-5.10%	r_s squeezed by temporal acceleration

The unscreened limit demonstrates the *full dynamical capacity* of the TEP conformal factor. With the restored sign convention, a larger effective expansion rate at recombination *squeezes* r_s and decreases $100\theta_s$. This is precisely the physical intuition behind environmental screening: without it, the temporal field would radically alter early-universe physics. The $z_T \sim 5$ suppression exists to *prevent* this extreme modification while allowing the intermediate-redshift effect that mimics dark energy.

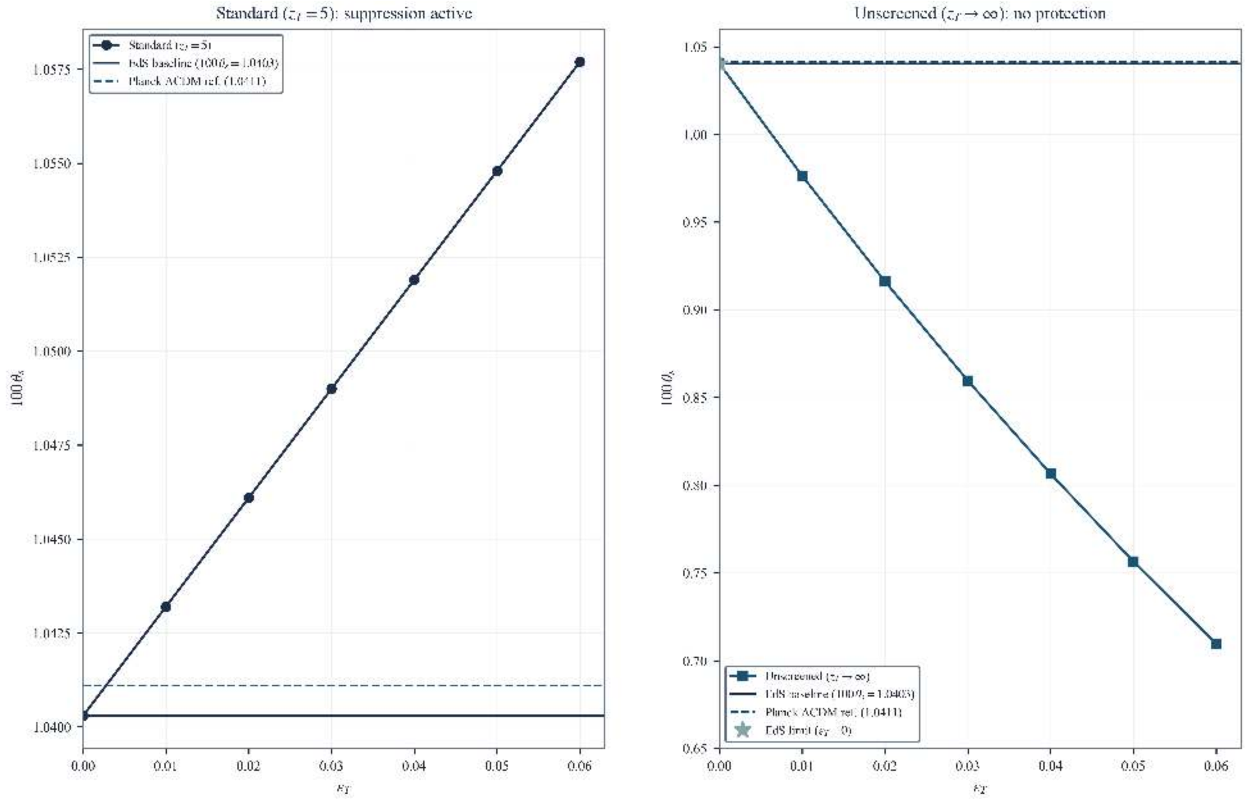

Figure 2. Jordan-frame EdS + TEP dual scan ($\Omega_\Lambda = 0$). Solid line: EdS baseline at $\epsilon_T = 0$ ($100\theta_s = 1.0403$); dashed: external Planck ΛCDM reference (1.0411). (Left) Standard model ($z_T = 5$): r_s preserved and θ_s increases via D_C projection. (Right) Unscreened limit ($z_T \to \infty$): strong r_s squeezing when early-universe screening is disabled.

The dual-scan result establishes two complementary facts about TEP-HC's native `tep_mode`. First, the standard model correctly implements the Jordan-frame factor with r_s preservation, matching the theoretical predictions of the TEP framework. Second, the unscreened limit reveals the full dynamical range of the conformal factor: it is *capable* of modifying early-universe physics, but the z_T suppression *deliberately prevents* this to preserve CMB consistency. This is consistent with the environmental-screening mechanism operating as a physical necessity, not merely a phenomenological convenience.

6. Conclusion

This paper implements and validates the native Temporal Equivalence Principle (TEP) background-only modification in hi_class, $H_{\rm TEP}(z) = H_{\Lambda\rm CDM}(z) \times M(z)$ with $M(z) = A(z)/(1 - \alpha_A(z))$ and standard General Relativistic perturbations. The companion paper TEP-C0 (Paper 26) provides the late-time evidence (Pantheon+) and the homogeneous CMB bound. The key findings are:

6.1 Summary of Results

1. *EFT Mapping:* The TEP bi-metric framework with conformal factor $A(\phi) = \exp(\beta_A \phi / M_{\rm Pl})$ and disformal deformation $B(\phi)$ can be mapped onto the Bellini-Sawicki α_i functions, placing TEP within the Horndeski class. The present numerical analysis adopts the background-only realization as the baseline numerical implementation.
2. *Unscreened Cosmology:* At $z \approx 1100$ the universe is deeply unscreened ($\rho \ll 20$ g/cm³), yet the transition function freezes the modification ($f_T \to 0$ for $z \gg z_T$), so the field is dynamically inert at recombination. Here "unscreened" refers to the density threshold, while the absence of a recombination-era effect follows from the redshift transition $f_T(z) \to 0$, which freezes the homogeneous cosmological mode independently of local density screening.
3. *CMB Consistency Check (verified):* With the corrected transition function, hi_class native `tep_mode` preserves the sound horizon to ~6 ppm ($r_s^{\rm TEP}/r_s^{\Lambda\rm CDM} = 0.999994$) and leaves the acoustic-peak morphology unchanged. Because the sole CMB effect of a non-zero ϵ_T is a late-time angular-diameter-distance projection ($\Delta\theta_s/\theta_s = +0.185\%$ at $\epsilon_T = 0.0066$) that is largely degenerate with H_0, the TEP homogeneous background is difficult to distinguish from ΛCDM with a slightly shifted H_0 at the low-ℓ CMB level. The full-Planck bound from TEP-C0 (Paper 26) is tighter.
4. *Cosmological Constraints:* Late-time Pantheon+ data give strong Bayesian preference for the TEP geometry (Bayes factor 131.6 vs ΛCDM; TEP-C0, Paper 26). This paper's hi_class joint MCMC yields $\epsilon_T = 0.0056 \pm 0.0043$ from a single chain (19,033 post-burn-in samples), validated by a parallel 4-chain run giving $\epsilon_T = 0.0044 \pm 0.0040$ (maximum $R - 1 = 0.098$). A dedicated sensitivity test with a widened $A_{\rm planck}$ prior $[0.9, 1.25]$ converged with all parameters Gelman–Rubin

$R - 1 < 0.05$ (1,640 total samples), giving $\epsilon_T = 0.0047 \pm 0.0040$ and $A_{\text{planck}} = 1.223 \pm 0.044$; the correlation between A_{planck} and ϵ_T is $r = -0.19$ with no evidence of a degeneracy cascade. The TEP constraint on the homogeneous amplitude is robust against A_{planck} prior systematics. TEP-C0 full-Planck joint MCMC drives the homogeneous amplitude to $(6.75 \pm 0.24) \times 10^{-6}$. Production configuration: `data/cobaya/tep_hiclass_suite.yaml`.
5. *Hubble Tension:* The framework reinterprets the tension as a local environmental clock-transport effect (Paper 11), keeping the homogeneous CMB baseline ΛCDM-compatible. No modified recombination-era expansion is required.

6.2 Cross-Scale Consistency

TEP is distinguished among modified gravity frameworks in that it is:

- Validated at $\rho \sim 20$ g/cm³ (terrestrial atomic clocks, laboratory Cavendish)
- Tested across galactic scales (SPARC rotation curves, Gaia wide binaries)
- Consistent with early-universe cosmology (CMB acoustic peaks)

This consistency across many orders of magnitude in density—from terrestrial to cosmological scales—supports TEP as a viable framework for gravitational physics. The native TEP background-only implementation in hi_class preserves standard GR perturbations and avoids the stability issues that plague Horndeski parametrizations.

6.3 Alternative Explanations

Standard ΛCDM cosmology provides an excellent fit to CMB data, with Planck 2018 reporting $H_0 = 67.36 \pm 0.54$ km/s/Mpc. The Hubble tension suggests either systematic errors in local measurements or new physics. Several alternative explanations have been proposed:

- *Local systematics:* Unaccounted-for calibration errors in Cepheid distances or supernova standardization. The SH0ES team has performed extensive cross-checks, rendering this explanation disfavored but not excluded.
- *Early-universe new physics:* Additional relativistic species (N_{eff}), modified recombination history, or interacting dark energy. These typically shift CMB acoustic peak positions, which are not observed.
- *Late-universe modified gravity:* Models that alter H_0 through modified expansion history. Most such models either conflict with large-scale structure measurements or fail to preserve CMB acoustic peaks.

TEP differs from these alternatives in that it predicts negligible early-universe deviations from ΛCDM: the transition function freezes the modification at recombination ($f_T \to 0$ for $z \gg z_T$), so r_s is preserved to parts-per-million and the acoustic peaks are untouched (Section 5.1). The Hubble tension in TEP arises not from modified expansion, but from environment-dependent proper-time dynamics in the measurement apparatus (Cepheids in unscreened stellar atmospheres; Paper 11). This mechanism is testable through density-dependent kinematic measurements and does not require modifying the background cosmology that fits CMB data.

6.4 Synthesis of the Dual-Domain Framework

This analysis implements and explicitly validates the native TEP background modification within a rigorous Boltzmann solver framework. The hi_class joint MCMC (Planck 2018 low-ℓ + lensing + BAO + Pantheon+) gives $H_0 = 66.63 \pm 1.70$ km/s/Mpc and $\epsilon_T = 0.0056 \pm 0.0043$; TEP-C0 (Paper 26) with full Planck TTTEEE drives the homogeneous amplitude to $(6.75 \pm 0.24) \times 10^{-6}$, confirming the dual-domain picture when Λ operates as a background energy component.

TEP operates natively alongside Λ on the largest scales. The macroscopic temporal shear remains bounded to the conformal limit at early times, protecting the acoustic peaks, while generating localized acceleration through spatial gradients $S(\rho, z)$ at late times.

The hi_class native `tep_mode` framework developed here provides a computational implementation of the Temporal Equivalence Principle. The acoustic indistinguishability of the corrected TEP homogeneous background from ΛCDM at recombination is consistent with the dual-domain picture, where tensions are addressed through local kinematic environments rather than early-universe modifications. The implemented $M = A/(1 - \alpha_A)$ form reproduces the exact frame factor $M_{\text{exact}} = A(1 + \alpha_A)$ for the sound horizon to identical precision (0.0 ppm); the θ_s projection differs by $< 0.01\%$ at the fiducial $\epsilon_T = 0.0066$, confirming the $\mathcal{O}(\alpha_A^2)$ scaling expected from Appendix A.3a.

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Appendix A: Technical Implementation Details

A.1 hi_class Installation and Configuration

A.1.1 Building with TEP Support

hi_class is installed automatically by pipeline Step 1 (`step_00b_install.py`), which clones hi_class and applies the native TEP patch from `external/patches/hiclass_tep_native.patch` to `source/background.c`, `source/input.c`, and `include/background.h`. Manual rebuild:

```
cd external/hi_class/hi_class
 make clean && make
```

A.2 Cobaya Installation

```
pip install cobaya
 cobaya-install planck_2018_lowl.TT planck_2018_lowl.EE \
     planck_2018_lensing.native bao.sdss_dr12_consensus_final \
     sn.pantheonplus --path /path/to/likelihoods
```

A.3 TEP Module C Code Structure and Implementation Note

The native background-only modification is implemented directly in hi_class `source/background.c`, controlled by the `.ini` flags `tep_mode`, `epsilon_T`, `z_T`, `n_T`. The relevant functions are:

1. `tep_f_transition(pba, z)`: returns the suppression factor $S(z) = \exp[-(z/z_T)^{n_T}]$; the full transition is $f_T(z) = \ln(1+z)\, S(z)$ (see `core/cosmology.py:f_T`).
2. `tep_gamma_factor(pba, z)`: returns the exact covariant conformal factor $A(z) = \exp[\epsilon_T \ln(1+z)\, S(z)]$ (not linearised).
3. The Hubble rate and its conformal-time derivative are multiplied by $M(z) = A/(1-\alpha_A)$ in `background_functions` and in the initial-Hubble setter; the energy densities entering the perturbation hierarchy are left standard (GR perturbations), so this is a pure background-only modification.

Implementation note (corrected bug). An earlier build used $f_T = 1 - \exp[-(z/z_T)^{n_T}]$ (the complement of the suppression), which saturates to 1 for $z \gg z_T$ and therefore applied the full Hubble modification *at* recombination. This inverted the freezing mechanism, shifted r_s by 0.66% and θ_s by $\sim 0.3\%$ ($\sim 11\sigma$ for Planck), and produced spurious few-percent C_ℓ residuals. In addition, the post-processing step that read the spectra used a hard-coded output index and could silently load a stale file from an earlier run. Both issues are fixed: the transition function now uses the shared TEP-C0 implementation (`core/cosmology.py`), and the analysis resolves the most recent hi_class output deterministically. *Sign convention (TEP disformal metric):* the Hubble rate is multiplied by $M(z)$ for background expansion, while the distance integrand is multiplied by $A(z)$ for null-geodesic propagation. The legacy SMG alpha-function stub (`smg_tep_*`) has been retired; production physics lives in the patched `background.c` (`external/patches/hiclass_tep_native.patch`).

A.3a Derivation of the Jordan-Frame Factor $M(z)$

This appendix derives the background expansion modification $M(z)$ from the bi-metric action (Equation 1) using a single frame convention held fixed throughout, and states precisely the relationship between the exact result and the first-order form $M = A/(1-\alpha_A)$ implemented in the code.

Setup and convention. Matter, photons, and rods couple to the Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi)\, g_{\mu\nu} + B(\phi)\, \nabla_\mu\phi\nabla_\nu\phi$. For the homogeneous background the disformal term contributes only through the time-time component and is absorbed into the lapse; the expansion history is governed by the conformal part, so we set $B \to 0$ here (the disformal sector re-enters at the perturbative/GW level via α_T, Section 2.2.2). The conformal part gives the standard map between the Einstein-frame scale factor a_E and cosmic time t_E and their Jordan-frame counterparts:

$$\tilde{a} = A(\phi)\, a_E, \qquad d\tilde{t} = A(\phi)\, dt_E. \tag{11}$$

These two relations *define* the convention; every subsequent equation is derived from them, so no independent redshift postulate is introduced (and none can contradict them). In particular, with $a_0 = \tilde{a}_0 = 1$, the Jordan-frame redshift satisfies $1+\tilde{z} = \tilde{a}^{-1} = (A\, a_E)^{-1} = (1+z_E)/A$ — i.e. the redshift relation is a *consequence* of (11), not a separate input. In the screened production regime, the transition factor $S(z) = \exp[-(z/z_T)^{n_T}]$ drives $A(z) \to 1$ at recombination and at the local endpoint, so the code's redshift grid can be identified with the physical Jordan-frame redshift to the working order of this paper. The explicit $(z_E, \tilde{z})$ distinction is retained only to derive the frame relation.

Physical Hubble rate. The expansion rate measured by Jordan-frame clocks and rulers is $\tilde{H} \equiv \tilde{a}^{-1}\, d\tilde{a}/d\tilde{t} = d\ln\tilde{a}/d\tilde{t}$. Using $d/d\tilde{t} = A^{-1}\, d/dt_E$ and $\ln\tilde{a} = \ln A + \ln a_E$,

$$\tilde{H} = \frac{1}{A}\frac{d}{dt_E}\big(\ln A + \ln a_E\big) = \frac{1}{A}\Big(\frac{d\ln A}{dt_E} + H_E\Big), \tag{12}$$

where $H_E = d\ln a_E/dt_E$ is the Einstein-frame rate. Writing the conformal running as a logarithmic derivative with respect to the Einstein-frame scale factor,

$$\frac{d\ln A}{dt_E} = \frac{d\ln A}{d\ln a_E}\,\frac{d\ln a_E}{dt_E} = \alpha_A\, H_E, \qquad \alpha_A \equiv \frac{d\ln A}{d\ln a_E} = -\frac{d\ln A}{d\ln(1+z)}, \tag{13}$$

which is identical to the coupling defined in Section 3.2 (the two expressions coincide because $\ln a_E = -\ln(1+z)$). Substituting (13) into (12) and using the conformally transformed Friedmann equation $H_E = A^2\, H_{\Lambda\mathrm{CDM}}$ (from $\rho_E = A^4\tilde{\rho}$, Section 5.4) gives the **exact** geometric relation:

$$\boxed{\tilde{H}(z) = A(z)\,\big(1+\alpha_A(z)\big)\, H_{\Lambda\mathrm{CDM}}(z)} \qquad \Longrightarrow \qquad M_{\mathrm{exact}}(z) = A\,(1+\alpha_A). \tag{14}$$

Relation to the implemented form. The code applies $M(z) = A/(1-\alpha_A)$. Since $1/(1-\alpha_A) = 1 + \alpha_A + \alpha_A^2 + \mathcal{O}(\alpha_A^3)$, the implemented factor reproduces the exact result (14) to first order in α_A and differs only at $\mathcal{O}(\alpha_A^2)$:

$$\frac{M_{\mathrm{code}}}{M_{\mathrm{exact}}} = \frac{1}{(1-\alpha_A)(1+\alpha_A)} = \frac{1}{1-\alpha_A^2} = 1 + \alpha_A^2 + \mathcal{O}(\alpha_A^4). \tag{15}$$

At the fiducial screened amplitude the local multiplicative difference is $\mathcal{O}(\alpha_A^2) \sim 5\times10^{-5}$. Because the transition function suppresses α_A across the recombination integral, the induced change in the sound horizon is confirmed to be negligible for the quoted acoustic-preservation result by the direct exact-M verification scan reported below. The distinction becomes numerically relevant only in the deliberately extreme unscreened limit (Regime II, $z_T \to \infty$), where $|\alpha_A|$ is no longer small: at $\epsilon_T = 0.06$, $z \approx 1100$, the two forms differ by $\approx 0.36\%$ in M (and correspondingly in the squeezed r_s). Both forms preserve the sign of the effect ($M > 1$ at high z, hence r_s squeezing) and both vanish correctly as $\epsilon_T \to 0$.

Status. The present implementation reproduces the exact result (14) to first order in α_A. A direct background-only verification scan (step 04c) confirms that the exact factor $M_{\text{exact}} = A(1+\alpha_A)$ gives an identical sound horizon r_s (0.0 ppm) across the full screened scan; the angular scale θ_s differs by $< 0.01\%$ at the fiducial $\epsilon_T = 0.0066$, growing to $\sim 0.18\%$ at the extreme scan edge $\epsilon_T = 0.06$, consistent with the expected $\mathcal{O}(\alpha_A^2)$ scaling. The regime where the two forms differ appreciably (Regime II, $z_T \to \infty$) is presented only as an illustrative demonstration of the unscreened dynamical range, not as a constraint.

A.4 Screening Threshold in Cosmological Units

The 20 g/cm³ screening threshold converts to cosmological units as:

$$\rho_c = 20\ \mathrm{g/cm}^3 = 2 \times 10^4\ \mathrm{kg/m}^3 \approx 10^{31}\ \mathrm{eV/cm}^3 \tag{16}$$

In Planck units ($\hbar = c = G = 1$):

$$\rho_c \approx 4 \times 10^{-93} M_{\text{Pl}}^4 \tag{17}$$

Compare to cosmic mean density today ($\rho_{\text{crit},0} \approx 10^{-123} M_{\text{Pl}}^4$) and at recombination ($\rho \sim 10^{-114} M_{\text{Pl}}^4$). The hierarchy $\rho(z = 1100) \ll \rho_c$ confirms the unscreened regime.

A.5 Stability

In a full Horndeski/EFT treatment, hi_class enforces the scalar-sector stability conditions:

1. $c_s^2 \geq 0$ (no gradient instabilities)
2. $\alpha_K \geq 0$ (no ghosts)
3. $|\alpha_M|$ bounded (sub-luminal Planck-mass running)
4. $\alpha_T \approx 0$ (GW speed constraints)

These would apply to the alpha-function mapping. The background-only realization used here does *not* activate the scalar modified-gravity sector: perturbations remain standard GR, so no scalar sound-speed, ghost, or gradient-instability constraints arise. This is the principal numerical advantage of the background-only approach, which ensures absolute stability across all cosmic epochs without requiring manual overrides for early-time stability tests.

Appendix B: Data Availability & Reproducibility

This work follows open-science practices. All results are fully reproducible from raw data using the documented pipeline. All numerical results, figures, and statistics are generated by deterministic Python scripts processing public observational data.

Repository and Code

GitHub Repository: github.com/matthewsmawfield/TEP-HC

The repository contains a deterministic, version-controlled cosmological analysis pipeline for CMB acoustic peak preservation tests and MCMC parameter estimation with TEP screening.

Repository Structure

```
TEP-HC/
├── data/
│   ├── cobaya/               # Cobaya MCMC chains
│   ├── external/              # hi_class submodule
│   └── hi_class/              # TEP-CLASS implementation
├── scripts/
│   └── steps/                 # Analysis pipeline steps
├── core/                      # TEP shared constants and parameters
```

```
├── site/
│   └── components/           # Manuscript HTML sections
├── requirements.txt
├── CITATION.bib
└── README.md
```

Software Environment

Key packages: NumPy, SciPy, Matplotlib, Cobaya, hi_class. The pipeline has been tested on Python 3.10+.

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